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The Python implementation of the dictionary is faster than my implementation of a hash map. It is faster because it does not use linked lists or trees but relies on arrays. With better hashing and simple index mod implementation, in Python's case, it does not need to care too much about collisions, but if there is a collision, it uses a probe method, which makes the usual case to find the collision value in the near vicinity of the index. Also aggressively resize to maintain the speed, as the closer it gets to the near size-load speed of the hashmap, the slower it becomes, especially for sets looking for probing. While other languages like Java use linked lists for small values and then switch to red-black trees, which are trees that automatically balance themselves, operations of search, insert, and delete become O(log n), but in most cases Python's implementation of the dictionary will win with O(1) but in the worst case of O(n).